

Understand the Series & Set the Search Rule

Time Series: WTI Spot Price at Cushing (2024)



Metadata
Series: RWTC
Domain: Energy
Frequency: 1d
Target: wti_spot_price
unit: USD per barrel
provider: EIA
WTI spot price at Cushing, Oklahoma

Proposed Search Rules

- Coverage — Jan-Dec 2024**
Cover the complete observation period of the RWTC series.
- Review Windows — Quarterly**
Divide the year into Q1-Q4 and review each window.
- Event Scope — Four Categories**
production policy, physical supply and operations, geopolitical risks, and demand outlooks.
- Search Budget — Up to 2 per Quarter**
Allow additional searches only after the initial evidence has been reviewed

HUMAN GATE 1
Approve Search Plan

- Are the event types relevant?
- Are the quarterly intervals and search limits reasonable?
- Search Period: Jul-Sep 2024

Decision: APPROVE

Search, Check & Confirm Events

WINDOW-WISE SEARCH · REAL Q1 OPEC+ EXAMPLE

Q1 · JAN-MAR
EXPANDED BELOW

Q2 · APR-JUN

Q3 · JUL-SEP

Q4 · OCT-DEC
THEN REPEAT FOR Q2-Q4

1 QUARTER = 1 WINDOW → Candidate events → SEARCH → EVIDENCE REVIEW → REFINE IF NEEDED

1 LLM PROPOSES POSSIBLE Q1 EVENTS

- OPEC+ cut extension · around 3 Mar
- Red Sea shipping risk · Jan-Mar
- IEA demand outlook · Mar

2 LLM WRITE SEARCHE QUERIES

- March 3 2024 voluntary production adjustments extension second quarter official
- 2024 OPEC+ extends voluntary oil cuts Q2 supply balance independent analysis

3 CLI RUNS THE SEARCHES

Search official sources → 0 results

Independent news reports search → 2 useful reports

4 LLM READS & COMPARES THE EVIDENCE

- Reuters · 3 Mar: 2.2 million b/d of voluntary cuts extended through Q2
- Bloomberg · 3 Mar: about 2 million b/d held to mid-year

AGREE: same date and same supply decision

GAP: first-party page missing → Refine the query and search again

EVIDENCE INCOMPLETE → REVISE THE SEARCH

5 LLM IMPROVES THE QUERY FROM THE GAP

- Use exact dates supported
- Broaden beyond one website
- Combine unresolved evidence gaps

6 LLM VERITY THE FINAL EVENT

Date	Event	Decision
16 Jan · U.S. deep freeze	physical production and refinery disruption	Accept
12-29 Jan · Red Sea escalation	shipping risk; realized oil-flow loss remained limited	Reject
3 Mar · OPEC+ cut extension	same decision already supported	Accept
14 Mar · IEA demand outlook	forecast raised to 1.3 million b/d	Accept

HUMAN GATE 2
Approve Event

- Do all files and references agree?
- Do dates match the evidence?
- Are unsupported cause-and-effect claims avoided?

Decision: APPROVE

Release the Dataset

- Dataset**
what the dataset contains
- Series-Event Links**
series description + links to relevant events
- Unified Event**
dated, self-contained verified event summaries

publish